Elementary bounds on the sample variance

Abstract

We collect elementary facts about the sample mean and the sample variance of a finite set of real observations: two identities, an upper bound for observations in a bounded interval, Samuelson's inequality, and a bound in terms of the raw second moment.

1 Setting

Throughout, $n \geq 1$ is an integer and $x = (x_1, \ldots, x_n) \in \mathbb{R}^n$.

Definition 1.1. The *sample mean* of x is $\bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i$.

Definition 1.2. For $n \geq 2$, the *sample variance* of x is

$$s^2 = \frac{1}{n-1} \sum_{i=1}^{n} (x_i - \bar{x})^2,$$

and the *sample standard deviation* is $s = \sqrt{s^2} \geq 0$.

2 Results

Lemma 2.1. $\sum_{i=1}^{n} (x_i - \bar{x}) = 0$.

Proof. By the definition of $\bar{x}$, $\sum_{i=1}^{n} (x_i - \bar{x}) = \sum_{i=1}^{n} x_i - n\bar{x} = 0$. □

Lemma 2.2. $\sum_{i=1}^{n} (x_i - \bar{x})^2 = \sum_{i=1}^{n} x_i^2 - n\bar{x}^2$.

Proof. Expanding the square and using $\sum_{i=1}^{n} x_i = n\bar{x}$, we obtain the claim. □

Proposition 2.3. *Suppose that $n \geq 2$ and that $a \leq x_i \leq b$ for $i = 1, \ldots, n$. Then*

$$s^2 \leq \frac{n}{n-1} \cdot \frac{(b-a)^2}{4}.$$

Proof. Popoviciu's inequality on variances [1] states that $\frac{1}{n} \sum_{i=1}^{n} (x_i - \bar{x})^2 \leq \frac{(b-a)^2}{4}$ whenever $x_1, \ldots, x_n$ lie in $[a, b]$. Multiplying both sides by $\frac{n}{n-1}$ gives the claim. □

Proposition 2.4 (Samuelson's inequality)**.** *Suppose that $n \geq 2$. Then $|x_i - \bar{x}| \leq \frac{n-1}{\sqrt{n}}\, s$ for every $i = 1, \ldots, n$.*

Proof. Let $z_j = (x_j - \bar{x})/s$ for $j = 1, \ldots, n$. By Lemma 2.1, $\sum_{j=1}^n z_j = 0$, and by the definition of s^2, $\sum_{j=1}^n z_j^2 = n - 1$. Fix i. Then $z_i = -\sum_{j \neq i} z_j$, so by the Cauchy–Schwarz inequality

$$z_i^2 \leq (n-1) \sum_{j \neq i} z_j^2 = (n-1)\big(n - 1 - z_i^2\big).$$

Hence $nz_i^2 \leq (n-1)^2$, that is, $|z_i| \leq (n-1)/\sqrt{n}$. Multiplying by s gives the claim. □

Lemma 2.5. *Suppose that $n \geq 2$. Then $s^2 \leq \frac{1}{n} \sum_{i=1}^n x_i^2$.*

Proof. By Lemma 2.2, $(n-1)s^2 = \sum_{i=1}^n x_i^2 - n\bar{x}^2 \leq \sum_{i=1}^n x_i^2$. Dividing by n gives the claim. □

Corollary 2.6. *Suppose that $n \geq 2$ and that $|x_i| \leq 1$ for $i = 1, \ldots, n$. Then $s^2 \leq 2$.*

Proof. By Lemma 2.5, $s^2 \leq \frac{1}{n} \sum_{i=1}^n x_i^2 \leq 1 \leq 2$. □

References

[1] T. Popoviciu. Sur les équations algébriques ayant toutes leurs racines réelles. *Mathematica (Cluj)*, 9:129–145, 1935.